

# Acne

By Salsabiela Bani Hamad

# Introduction

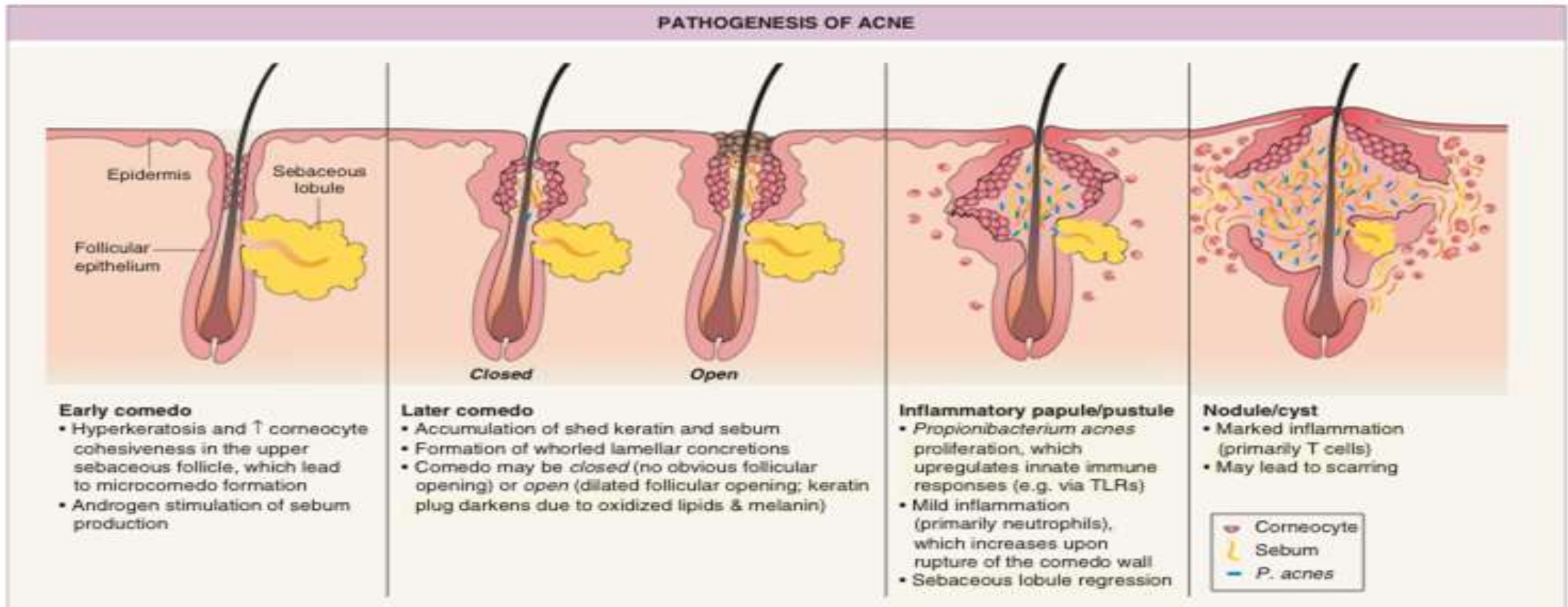
- Acne is a **multifactorial** disorder of the pilosebaceous unit
- All age groups may be affected, peak incidence is during adolescence.
- **Psychological impact**, social isolation, anxiety, depression even suicidal ideation.
- Severe acne may be very painful, cause irreversible scarring, and may be associated with systemic symptoms e.g. joint pain, fever, and fatigue.

# Acne

## **The main changes of acne are:**

1. Thickening of the keratin lining and subsequent obstruction of the sebaceous duct resulting in closed comedones ('whiteheads') or open comedones ('blackheads' whose color is due to melanin, not dirt). Acne cannot be diagnosed without the presence of comedones (**there are no comedones seen in rosacea**).
2. An increase in sebum secretion.
3. An increase in *Propionibacterium acnes* bacteria within the duct.
4. Inflammation around the sebaceous gland.

# Pathogenesis:



**Fig. 36.1** Pathogenesis of acne.

# Underlying factors:

- **Genetic factors:**

- Affect the number, size, and activity of sebaceous glands.
- Also, the tendency of severe acne runs in families.

- **Androgenic hormones:**

- Increase the size of sebaceous glands and sebum amount.
- Usually, androgen levels are normal but there is an increased sensitivity of the glands to androgens.

- **Dietary factors:**

- Controversial
- E.g. Skim milk, whey protein, high glycemic diets, dietary supplements vitamin B12, dairy products, fizzy drinks, and chocolates.

# Underlying factors:

- **Fluid retention:**

- Lead to increased hydration and swelling of sebaceous glands e.g., (premenstrual)
- Sweating also makes acne worse.

- **Stress:**

- Induces inflammation in the pilosebaceous unit.

- **Seasons:**

- Acne improves with natural sunlight, also phototherapy!!

# Underlying factors:

- **Iatrogenic:**

- Corticosteroids (topical can cause perioral dermatitis), Androgens, OCP, Antiepileptics, EGFR inhibitor.

- **External Factors:**

- Oil e.g., cooking, engineering
- Coal tar
- Cosmetics: also comedogenic.

((Pomade Acne: acne on the hair line as a result of the use of rich oils in the scalp))

# Perioral dermatitis:

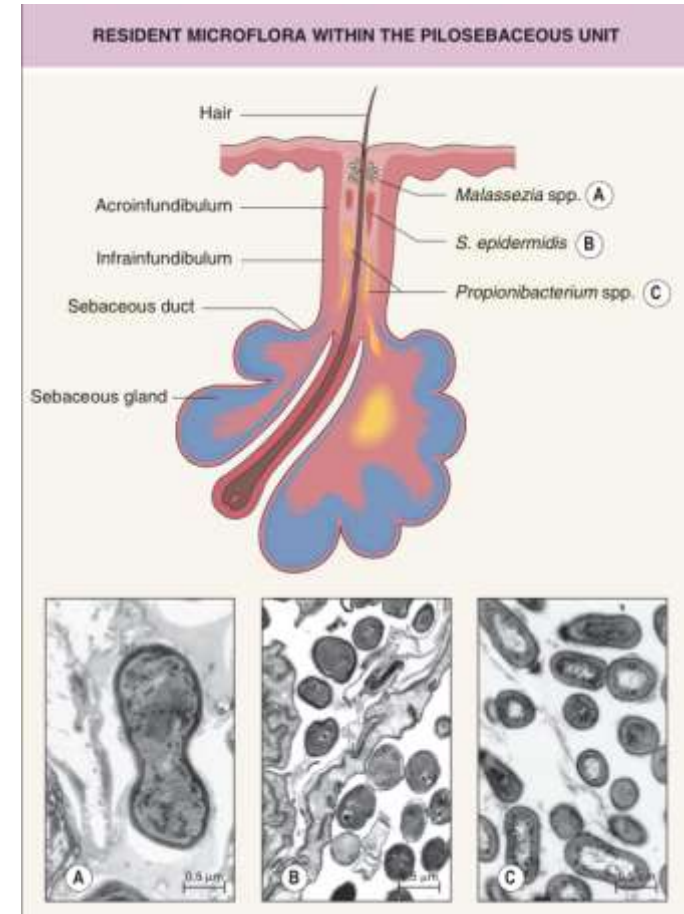
- After application of topical steroids to the **face**
- Characterized by **papules** and **pustules** around the mouth and eyes that may mimic acne **but there are no comedones**.
- The topical steroid should be stopped and the patient treated with **oral tetracycline** or **erythromycin** for 6 weeks.





## Pilosebaceous unit:

- Follicular hyper-keratinization corneocytes normally extruded in acne accumulate, hyperkeratotic plug.
- Hormonal influences on sebum production and composition of **testosterone** and **dihydrotestosterone** most potent.
- Inflammation and increased concentration of **Propionibacterium acnes** which uses sebum as a substrate, and produces lipases that break down the triglycerides in sebum to form free fatty acids.



## Box 12.1 Factors causing acne

### **Intrinsic factors**

#### *Hormones*

- Polycystic ovary syndrome.
- Virilising tumours.
- Congenital adrenal hyperplasia.
- Increased cortisol (Cushing's syndrome).
- Increased growth hormone (acromegaly).

#### *Medications*

- Topical and systemic steroids.
- Oral contraceptive pill (higher androgen content).
- Phenytoin.
- Barbiturates.
- Isoniazid.
- Ciclosporin.
- Lithium.

### **Extrinsic factors**

- Oils/pomades.
- Coal and tar.
- Chlorinated phenols.
- DDT and weed killers.

# Clinical features:

- Localization: sites of well-developed sebaceous glands.
- Non-inflammatory vs inflammatory
- Non-inflammatory: Open vs closed comedones
  - **Closed comedones** (white heads) small papules, no follicular opening, no erythema
  - **Open comedones** (blackheads) dilated follicular opening with core of shed keratin (color due to melanin and lipid oxidation).





**Black comedones  
(opened)**



**White comedones  
(Closed)**



**Fig. 36.3 Comedonal acne vulgaris.** On the cheek (A) and forehead (B), there are open and closed comedones as well as postinflammatory hyperpigmentation (A) and inflammatory papules (B). B, Courtesy, Kalman Watsky, MD



# Clinical features:

## Inflammatory acne:

Papules, pustules, nodules, pseudocysts (filled with pus and serosanguinous fluid) may combine to sinus tracts .



Fig. 36.5 Moderate to severe acne vulgaris. Inflammatory papules and pustules as well as both open and closed comedones are evident on the cheek, forehead and chin. Note the nodular lesion on the temple and open comedones in the concha of the ear. Scarring is present in the preauricular area.



Fig. 36.6 Moderate to severe acne vulgaris. Multiple coalescing papules, pustules, and small nodules are present on the cheek. Courtesy: Kalman Worsky, MD

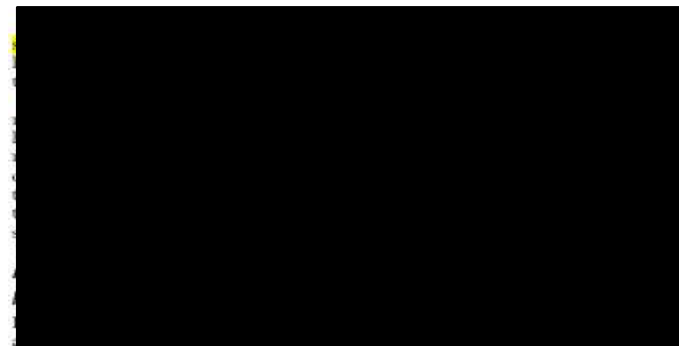


Fig. 36.7 Severe nodular acne. This form is best treated with low doses of isotretinoin initially, to avoid precipitating a flare.



Fig. 36.8 Postinflammatory hyperpigmentation secondary to acne. Such pigmentary changes are most common in patients with darker skin colors.



Fig. 36.9 "Ice-pick" scarring secondary to acne.

# Types of Acne:

## 1- Acne vulgaris:

- Acne vulgaris, the common type of acne, occurs during puberty and affects the comedogenic areas of the **face, back, and chest**.
- There may be a familial tendency to acne.
- Acne vulgaris is more common in boys, 30–40% of whom develop acne between the ages of 18 and 19.
- In girls, the peak incidence is between 16 and 18 years.

### Adult acne :

>> is a variant affecting 3% of men and 5% of women over the age of 40

### Acne Kerlroidalis:

>> is a type of scarring acne seen particularly on the neck in men



## 2. Infantile Acne:

- Localized acne lesions occur on the face in the first few months of life.
- Infantile acne may require topical or systemic therapy as although it will resolve spontaneously it may last up to 5 years and can cause scarring.
- There is an **association with severe adolescent acne**.
- Endocrine abnormalities are very rare but should be considered, especially in a girl with signs of virilism (e.g. precocious puberty)





### 3. Acne conglobata/fulminans:

- Acne conglobata is a severe form of acne, more common in boys and tropical climates. There is extensive, nodulocystic acne and abscess formation affecting particularly the trunk, face, and limbs.
- Acne fulminans is similarly severe but is associated with systemic symptoms of malaise, fever, and joint pains





**4. Occupational Acne.**

**5. Drug-Induced Acne.**

**6. Cosmetic-induced Acne.**

**7- Pyoderma faciale:**

- Produces erythematous and necrotic lesions and occurs mainly in adult women.

**8- Gram-negative folliculitis:**

- Occurs with a proliferation of organisms such as Klebsiella, Proteus, Pseudomonas, and Escherichia coli.



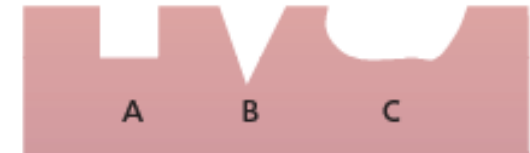
## 9.Acne excoriée:

- Typically seen in teenage girls who present with facial excoriations but very few primary lesions.
- The patient picks at the skin producing disfiguring erosions.
- The acne itself is usually mild but tends to be persistent as it is often very difficult to help the patient break this habit.
- There are **no comedones**.
- This is a form of **neurotic excoriation**, and patients need to be given a clear explanation and helped to try and reduce the self - inflicted damage.



# Acne complications:

- Resolving lesions leave post-inflammatory pigment changes and scarring.
- Scars may be atrophic and pitted, deep, (‘ice pick’), rolling, and boxcar or they may be more hypertrophic or even keloid.
- Scars may also be hyper/hypopigmented and erythematous.
- Treatment of acne scarring does depend on the severity and scar type



**Figure 12.7** Types of atrophic acne scars. (Notes: A, Boxcar; B, icepick and C, rolling).

# Treatment of acne: (cleansers, topical, systemic)

- Full explanation and active therapy are essential.
- Avoid precipitating factors
- Early intervention can also help reduce the likelihood of permanent scars and post-inflammatory pigment changes.
- Treatment may need to continue for months or years.
- When choosing a topical formulation to prescribe for a patient with acne it is worth bearing in mind the patient's skin type:
  - For dry/sensitive skin >> prescribe creams.
  - For oily skin >> use solutions or gels.
  - For combination skin and hair-bearing sites >> use lotions.

# Topical treatments:

1. Benzoyl peroxide: has been available for the treatment of acne for many years; it has **bacteriostatic** effects against *P. acnes* and is mildly **comedolytic**.  
S/E: irritation, & bleaching.

2. Salicylic acid: promotes the **desquamation** of follicular epithelium and therefore inhibits the formation of comedones.

3. Azelaic acid: has anti-keratinizing and antibacterial effects.  
S/E Mild skin irritation and depigmentation of the skin.



# Topical treatments:

**4. Topical retinoids:** are vitamin A derivatives that are **anti-inflammatory** and **comedolytic**. Treatments currently available include tretinoin and adapalene.

S/E: skin irritation, erythema, desquamation, initial acne flare, photosensitivity (so should be applied at night)

**5. Topical antibiotics:** are effective through their **bactericidal** activity against *P. acnes* and consequent **anti-inflammatory** effects. The most commonly prescribed antibiotics include erythromycin and clindamycin either alone or in combination with other agents such as zinc or benzoyl peroxide.

S/E: bacterial resistance.

**6. Phototherapy with ultraviolet or visible light:** this is an alternative therapy that may be helpful in those unresponsive to or unable to tolerate conventional treatments.

# Systemic therapy

- Oral antibiotics.
- Hormonal treatments.
- Oral retinoids.

# Hormone therapies:

- These include certain types of OCPs that increase sex hormone-binding globulin and consequently reduce free testosterone levels.
- These are generally OCPs that have higher estrogen and lower androgen potential.
- Antiandrogen treatment alone can be teratogenic e.g. spironolactone.

# Oral antibiotics:

- **Tetracyclines remain the mainstay of treatment** in those over the age of **12 years** (below this age, they may cause dental hypoplasia and staining of teeth)
- Tetracyclines should be avoided in pregnancy/breastfeeding and children <12 years.
- **Erythromycin** and **trimethoprim** are good alternatives.
- An adequate treatment course should last approximately **6–12 months**, depending on severity and response.

# Oral retinoids:

- **Isotretinoin is usually reserved for resistant diseases unresponsive to other oral therapies.**
- **Side effects:**
  - **Teratogenesis** (90% risk of birth defects), Female patients of childbearing age will need to use a robust form of contraception while taking isotretinoin and for 1 month following its cessation. Pregnancy testing is usually undertaken before the release of prescriptions for females.
  - **Rise in liver enzymes and lipids**, Blood testing before initiation is essential and during therapy as indicated clinically.
  - All patients will experience some **drying of the lips and skin.**
  - **Mood change and depression** (patient should be told).



# Oral retinoids:

- Isotretinoin dosing is to begin patients on a low dose for the first 1–2 months (20–40mg daily) and then increase to 1mg/kg/day to minimize initial xerosis.
- The cumulative target dosage for isotretinoin is **120–150mg/kg**.
- Occasionally, patients require a second or even third course of isotretinoin treatment.

**Table 12.1** Treatment of acne.

| Treatment   | Comedones                                          | Inflammatory papules/pustules                                                                           | Mixed picture                                                                                   | Nodulocystic                                                                                             |
|-------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| First line  | Topical retinoid<br>Azelaic acid<br>Salicylic acid | Benzoyl peroxide                                                                                        | Topical retinoid $\pm$ topical<br>antibiotic $\pm$ benzoyl peroxide<br>Combination of all three | Oral antibiotic + topical retinoid                                                                       |
| Second line | Physical comedone<br>extraction                    | Oral antibiotic<br>Oral contraceptive pill (high oestrogen,<br>low androgen, e.g. Yasmin <sup>®</sup> ) | Azelaic acid + benzoyl<br>peroxide $\pm$ topical antibiotic                                     | Oral isotretinoin<br>A short course of systemic steroids may<br>be given initially with the isotretinoin |
| Third line  |                                                    | Anti-androgens e.g. co-cypindiol<br>(Dianette <sup>®</sup> )<br>Different oral antibiotic               | Hormone therapy<br>Oral antibiotic<br>Oral isotretinoin                                         | Triamcinolone injections to unresponsive<br>lesions                                                      |

# Rosacea

# Introduction:

- Characterized by key components (Facial flushing, persistent facial erythema, telangiectasia, and inflammatory papules/pustules and edema).
- **Facial flushing and erythema are frequently exacerbated by:**
  - Heat
  - Exercise
  - Hot food/drinks
  - Spicy food
  - Emotion
  - Alcohol
  - Sunlight.



## Clinical types:

1. **Erythematotelangiectatic:** flushing, facial erythema +/- telangiectasia
2. **papulopustular:** centropacial eruption multiple small papules
3. **Phymatous:** thickened hypertrophic nodular skin, prominent pores, can affect nose (rhinophyma), chin, forehead, ears, eyelids
4. **Ocular:** dry gritty, eyelid edema, blepharitis, conjunctivitis.





**Fig. 37.2 Erythematotelangiectatic rosacea.** Persistent erythema of the medial and lateral cheeks is seen. In this patient, there are no telangiectasias, indicating mild (grade 1) disease.



**Fig. 37.3 Moderate papulopustular rosacea of the forehead.** Note the superficial nature of the inflammatory lesions.



**Fig. 37.4 Moderate to severe papulopustular rosacea.** There is a typical centrofacial distribution of erythema, telangiectasias, papules and pustules. In addition, the skin has a scaly, crusty surface and this is often a sign of more severe disease.





**Fig. 37.6 Rhinophyma.** Hypertrophy of sebaceous glands and connective tissue as well as patulous follicles are seen. The changes are more prominent in the mid to lower nose. In addition, there is evidence of papulopustular rosacea.

*Courtesy: Kathryn Watsky, MD*



**Fig. 37.9 Ocular rosacea.** **A** Tiny concretions of keratin (conical dandruff) are visible at the bases of some of the eyelashes of the lower eyelid. There is also evidence of blepharitis of the lower eyelid and conjunctival injection. **B** Erythema of the mucosal portion of the lower eyelid and ectropion. **C** Marked injection of the conjunctivae, leading to the appearance of red eyes. Ectropion is also present.

# Differential diagnosis of rosacea:

- **Acne**, in which there are comedones, a wider distribution, and improvement with sunlight.
- **Seborrhoeic eczema**, in which there are no pustules and eczematous changes are present.
- **Perioral dermatitis**, which occurs in women with pustules and erythema around the mouth and chin. This may be precipitated by the use of potent topical steroids.



# Differential diagnosis of rosacea:

- **Dermatomyositis**, where there may be periorbital and upper eyelid swelling and erythema.
- **Lupus erythematosus**, which shows light sensitivity, erythema, and scarring, but **no pustules**.



# Management:

Avoid trigger factors.

Sun screen

Avoid skin irritants

Topical treatment :Metronidazol, azelic acid, a adrenoreceptors  
( temporary effect).

Oral antibiotics

Low dose isotretinoin

Laser, surgery



# Management:

## Box 12.2 Ladder of treatment for rosacea

### Avoid exacerbators:

- Avoid harsh soaps, strong sunlight, oil-based make-up and hot showers.
- Avoid alcohol and spicy food.

+ Use Broad spectrum sunscreen regularly

### Topical treatment:

- Water-based emollient.
- Metronidazole gel or cream.
- Azelaic acid.
- Tacrolimus/pimecrolimus cream.

### Systemic treatment:

- Tetracycline antibiotics (minocycline 100 mg MR once daily (OD), lymecycline 408 mg OD,
- Low dose Isotretinoin.

- Topical preparations seem to be more effective at treating the papules and pustules than the erythema and flushing.
- However,  **$\alpha$ -adrenoreceptor agonists** have recently been evaluated for the treatment of diffuse facial erythema e.g. Oxymetazoline/xylometazoline, and brimonidine tartrate

- Laser ablation of dilated telangiectatic vessels with **a pulsed dye laser** can be undertaken once the inflammatory component has been treated.
- **Carbon dioxide laser** or **shave removal** with a scalpel blade of excess skin from the nose can significantly improve the appearance of rhinophyma.

Thank you